

CLIENT DELIVERABLE

SAOS Architecture Overview v4.0

Systack Documentation

ROLE	COLOR	HEX
Headers, CTAs	Navy	 #001a2d
Navy Light	Navy Light	 #002845
Secondary accents	Teal	 #007da9
Primary buttons, links	Cyan	 #00a1db
Gradients	Cyan Bright	 #00c5e0
Backgrounds	Gray 50	 #f8fafc
Cards	Gray 100	 #f1f5f9
Borders	Gray 200	 #e2e8f0
Muted text	Gray 400	 #94a3b8
Body text	Gray 600	 #475569
Headings	Gray 800	 #1e293b
Success	Green	 #22c55e
Error	Red	 #ef4444
Accent highlights	Purple	 #8b5cf6

SAOS Architecture Overview

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SAOS System Architecture

Overview

SAOS (Systack Agent Orchestration System) is a 10-agent AI fleet deployed on dedicated VPS instances, orchestrated via PostgreSQL task queues and accessible through Tailscale-secured networks. The customer portal (v2.1) provides PIN-based authentication, real-time chat, task tracking, and service management.

1. Fleet Composition

10-Agent Canonical Fleet

TIER	AGENT	EMOJI	ROLE	DESCRIPTION	CAPABILITIES
Execution	SOL	👤	System Operations Liaison	Main point of contact. Coordinates fleet, handles strategic requests.	system-optimization, strategic-planning, fleet-coordination, automation-design
Execution	CODY	💻	Code Agent	Writes, reviews, deploys code. Website changes, API integrations.	code-generation, deployment, technical-reviews, bug-fixes
Execution	ASSEMBLY	🔧	Integration Builder	Connects tools. Workflows between Stripe, Slack, email, databases.	workflow-automation, integration-setup, api-connections, data-pipelines
Quality	VALI	✅	Validation Agent	Quality control. Reviews code, tests automations, verifies data.	code-review, testing, validation, quality-assurance
Quality	PESSI	⚠️	Pessimist / Risk Agent	Identifies edge cases, security risks, failure modes.	risk-analysis, edge-case-detection, security-review, failure-mode-analysis
Intelligence	ORACLE	🧠	Research Agent	Deep research, analysis, strategic evaluation.	market-research, technology-evaluation, strategic-analysis, competitive-intelligence
Intelligence	ATLAS	🌐	Deployment Agent	Infrastructure and DevOps. Servers, Tailscale, system health.	infrastructure, deployment, devops, server-management
Engagement	CHATTY	💬	Communication Agent	Messaging, outreach, content creation. Drafts emails, social posts.	content-creation, email-drafting, social-media, customer-communications
Engagement	GENI	🎨	Creative Agent	Images, videos, creative assets. Marketing visuals and multimedia.	image-generation, video-generation, creative-design, brand-assets
Compliance	JURIS	⚖️	Legal & Compliance Agent	Reviews for compliance, audits configs, ensures legal requirements.	compliance-review, legal-audit, policy-check, deployment-review

System Loop (Canonical)

ORACLE → Design → CODY → Build → ASSEMBLY → Deploy → VALI → Validate → PESSI → Stress-test → SOL → Execution

2. Infrastructure

VPS Tiers

PLAN	VULTR TIER	VCPU	RAM	STORAGE	MONTHLY COST
Business	vhp-8c-16gb-amd	8	16GB	160GB NVMe	\$96
Enterprise	voc-g-8c-32gb-160s-amd	8	32GB	160GB NVMe	\$240

Network Architecture

```
Client Device (Tailscale)
  ↓
Tailscale Tailnet (Encrypted Mesh)
  ↓
VPS (Ubuntu 22.04 + Tailscale)
  ├── Port 8765: Fleet Dashboard API
  ├── Port 8766: Invoice Dashboard API
  ├── Port 8768: Customer Portal API (v2.1)
  ├── Port 18789: OpenClaw Gateway
  └── PostgreSQL (systack_memory)
```

Persistence

SERVICE	PORT	LAUNCHAGENT	STATUS
Fleet Dashboard	8765	net.systack.dashboard	✓ Running
Invoice Dashboard	8766	net.systack.invoice-dashboard	✓ Running
Customer Portal	8768	net.systack.customer-dashboard	✓ Running
Orchestrator	N/A	net.systack.orchestrator	✓ Running
Webhook Bridge	8767	net.systack.webhook-bridge	✓ Running

3. Customer Portal v2.1 — Architecture

Overview

The customer portal is a single-page application served by a Flask API. It provides PIN-based authentication, real-time chat, task tracking, and service management for SAOS clients.

Component Diagram

```
Client Browser
  ↓ (HTTPS via Tailscale)
Tailscale Serve (reverse proxy /dashboard/)
  ↓
Flask API (api.py, port 8768)
  ├── Static file serving (index.html)
  ├── Auth endpoints (/api/auth/*)
  ├── Portal endpoints (/api/portal/*)
  └── Chat endpoints (/api/chat/*)
```

- └ PDF downloads (/download/*)
- └ PostgreSQL (systack_memory DB)

Authentication Flow

ONBOARDING FLOW

```
Admin generates temp PIN (admin_client.py)
↓
Client visits dashboard → "First time? Set up your PIN"
↓
Client enters: Client ID + Temp PIN + New PIN
↓
POST /api/auth/register → validates temp PIN
↓
Sets auth_pin, onboarding_status='active', clears temp_pin
↓
Auto-login with new PIN
```

REGULAR LOGIN FLOW

```
Client enters: Client ID + PIN
↓
POST /api/auth/login → validates PIN against saos_clients
↓
Generates session token (SHA-256 hashed, 30-day expiry)
↓
Token stored in localStorage, sent as Bearer header
↓
All API calls require: Authorization: Bearer <token>
```

API Endpoints (v2.1)

ENDPOINT	METHOD	AUTH	PURPOSE
/api/portal/health	GET	None	Health check
/api/portal/onboarding-status/<id>	GET	None	Check if client needs PIN setup
/api/auth/login	POST	None	Login with Client ID + PIN
/api/auth/register	POST	None	First-time PIN setup from temp PIN
/api/auth/forgot-pin	POST	None	Request PIN reset via email
/api/auth/change-pin	POST	Bearer	Change PIN (logged-in clients)
/api/auth/logout	POST	Bearer	Revoke current token
/api/auth/me	GET	Bearer	Get current client info

ENDPOINT	METHOD	AUTH	PURPOSE
/api/portal/status	GET	Bearer	Fleet overview, metrics, agents
/api/portal/tasks	GET	Bearer	Task list with descriptions
/api/portal/agents	GET	None	Agent list with roles + descriptions
/api/portal/client	GET	Bearer	Client info
/api/chat/conversations	GET	Bearer	List chat conversations
/api/chat/conversations	POST	Bearer	Create new conversation
/api/chat/conversations/<id>/messages	GET	Bearer	Get messages
/api/chat/conversations/<id>/messages	POST	Bearer	Send message (auto-creates task)
/api/chat/conversations/<id>/close	POST	Bearer	Close conversation
/api/chat/webhook/agent-response	POST	None	Agent → chat (n8n/OpenClaw webhook)
/api/chat/poll	GET	Bearer	Poll for new messages + task updates
/download/*	GET	None	PDF document downloads

Database Schema (v2.1 Additions)

```

-- Task queue enhancements
ALTER TABLE task_queue ADD COLUMN display_name VARCHAR(255);
ALTER TABLE task_queue ADD COLUMN description TEXT;

-- Agent state enhancements
ALTER TABLE agent_state ADD COLUMN role VARCHAR(255);
ALTER TABLE agent_state ADD COLUMN role_description TEXT;
ALTER TABLE agent_state ADD COLUMN capabilities JSONB DEFAULT '[]';
ALTER TABLE agent_state ADD COLUMN avatar_emoji VARCHAR(10);
ALTER TABLE agent_state ADD COLUMN tier_access VARCHAR(50) DEFAULT 'all';

-- Client onboarding
ALTER TABLE saos_clients ADD COLUMN onboarding_status VARCHAR(50);
ALTER TABLE saos_clients ADD COLUMN temp_pin VARCHAR(10);
ALTER TABLE saos_clients ADD COLUMN temp_pin_expires_at TIMESTAMP;
ALTER TABLE saos_clients ADD COLUMN last_login_at TIMESTAMP;
ALTER TABLE saos_clients ADD COLUMN login_count INTEGER DEFAULT 0;

-- Client invitations
CREATE TABLE client_invitations (
  id SERIAL PRIMARY KEY,
  client_id INTEGER REFERENCES saos_clients(id),
  invite_code VARCHAR(64) UNIQUE,
  email VARCHAR(255),
  expires_at TIMESTAMP NOT NULL,

```

```
used_at TIMESTAMP,  
created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP  
);
```

Frontend Architecture

COMPONENT	TECHNOLOGY	NOTES
SPA	Vanilla HTML/CSS/JS	No framework dependency
Auth	Bearer token in localStorage	30-day expiry, SHA-256 hashed
Chat polling	setInterval 5s	Real-time message updates
Mobile	Responsive CSS + hamburger nav	Touch-optimized, iOS safe areas
Modal system	CSS overlay + JS	Task detail + agent detail modals

Admin Tools

```
# Generate temp PIN for new client  
python3 admin_client.py --create-temp-pin <client_id>  
  
# List all clients  
python3 admin_client.py --list-clients  
  
# Check specific client status  
python3 admin_client.py --client-status <client_id>
```

4. Service Tiers

TIER	PRICE	TARGET	KEY FEATURES
Personal	\$49/mo	Individuals	Email triage, calendar, reminders, research
Personal+	\$99/mo	Power users	+ Voice, multi-device, expense tracking
Business	\$299/mo	SMBs	Invoice processing, lead qualification, team chat
Enterprise	\$799/mo	Large orgs	On-premise, HIPAA, white-glove, 4hr SLA
Private	\$799/mo	Regulated	Air-gapped, zero cloud, audit trail
Accelerate	\$249/mo	Startups	Cloud GPU, auto-scaling, fast deployment

5. Security Model

Network Security

- **Tailscale WireGuard** encryption for all traffic
- Tailnet-only access (no public internet exposure)
- HTTPS via Tailscale TLS termination

Application Security

- PIN-based authentication (4–10 digits)
- Session tokens hashed with SHA-256
- 30-day token expiry with revocation
- Client-scoped data access (all queries filter by client_id)

Data Security

- PostgreSQL stored locally on VPS
- No cloud data exposure (except Stripe — client’s own account)
- All AI processing on local Ollama models

6. Dashboard Tabs (v2.1)

TAB	PURPOSE	KEY FEATURES
Chat	Client ↔ Agent messaging	Conversation sidebar, auto-task creation, polling
Dashboard	Fleet overview	Metrics, agent cards with descriptions + capabilities
Services	Plan-specific services	Tier-based service list, infrastructure, support info
Tasks	Task management	Filterable list, clickable detail modal, descriptions
Activity	Recent activity	Last 20 tasks as feed, clickable for details
Docs	PDF downloads	Tier-specific document access
Settings	Account management	Change PIN, view account info, deployment details

Internal document. Not for client distribution.